

Bihar Engineering University, Patna

B.Tech 2nd Semester Examination, 2025

Course: B.Tech
Code: 100209

Subject: Basic Electronics

Time: 03 Hours
Full Marks: 70

Instructions:-

- (i) The marks are indicated in the right-hand margin.
- (ii) There are **NINE** questions in this paper.
- (iii) Attempt **FIVE** questions in all.
- (iv) Question No. 1 is compulsory.

Q.1 Choose the correct option / answer the following (Any seven question only):

[2 x 7 = 14]

- (a) When a pure semiconductor is heated, its resistance will
 - (i) Goes up
 - (ii) Remains the same
 - ~~(iii)~~ Goes down
 - (iv) None
- (b) The random motion of holes and free electrons due to thermal agitation is called
 - (i) Diffusion
 - ☒ (ii) Ionisation
 - (iii) Pressure
 - (iv) None of these
- (c) A zener diode is used as
 - (i) An amplifier
 - (ii) A rectifier
 - ~~(iii)~~ A voltage regulator
 - (iv) A multivibrator
- (d) The ripple factor of a half-wave rectifier is
 - (i) 2
 - (ii) 0.48
 - ~~(iii)~~ 1.21
 - (iv) 2.5
- (e) The number of depletion layers in a transistor is
 - (i) Four
 - (ii) Two
 - ~~(iii)~~ Three
 - (iv) One
- (f) In a transistor, $I_C = 100 \text{ mA}$ and $I_E = 100.2 \text{ mA}$. The value of β is
 - (i) 100
 - (ii) 50
 - ~~(iii)~~ 1
 - (iv) 200
- (g) A non inverting closed loop op-amp circuit generally has a gain factor
 - (i) Less than one
 - (ii) Zero
 - ~~(iii)~~ Greater than one
 - (iv) Equal to one
- (h) If ground is applied to the (+) terminal of an inverting op-amp, the (-) terminal will
 - (i) Not need an input resistor
 - ☒ (ii) Have high reverse current
 - (iii) Be virtual ground
 - (iv) Not invert the signal
- (i) The binary code of $(21.125)_{10}$ is
 - ☒ (i) 10101.001
 - (ii) 10101.010
 - (iii) 10100.001
 - (iv) 10100.111
- (j) Which of the following gate is a two-level logic gate
 - (i) OR gate
 - (ii) EXCLUSIVE OR gate
 - ~~(iii)~~ NAND gate
 - (iv) NOT gate

- Q.2 (a) Explain the working of p-n junction diode and draw its V-I Characteristics. [7]
(b) Draw a neat circuit diagram of bridge rectifier and explain its operation with output waveforms. Drive the average value of current and voltage. [7]

- Q.3** (a) Explain working principle of Depletion type MOSFET (n-channel). Draw & Explain its characteristics. [7]
 (b) Explain the working principle and characteristics of a full-wave rectifier. Derive the expressions for output voltage and ripple factor for a full-wave rectifier. [7]
- Q.4** (a) Draw the block diagram of Op-Amp and list all the ideal characteristics of op-amp. [7]
 (b) Explain the working of op-amp as an Integrator and derive its output equation. [7]
- Q.5** (a) Draw the circuit of NPN transistor in common base configuration and discuss its working. Draw input-output characteristic. [7]
 (b) Explain the output and transfer characteristics of a JFET. How does the gate-source voltage (V_{gs}) affect the drain current (I_d)? [7]
- Q.6** (a) Convert the following: [7]
 (i) $(53.625)_{10}$ to $(?)_2$
 (ii) Find the base x if $(211)_x = (152)_8$
 (iii) Subtract using 1's complement: $(10111)_2 - (110011)_2$
 (iv) Find the 1's and 2's complement of $(010100)_2$
 (b) Discuss the working of basic logic gates: NAND, NOR, XOR. Derive the truth tables and Boolean expressions for each gate. [7]
- Q.7** (a) Explain the concept of load line analysis in a BJT. Derive the expressions for the load line and operating point. Explain their significance in amplifier design. [7]
 (b) Explain the difference between intrinsic and extrinsic semiconductors. How does the addition of impurities affect the electrical properties of semiconductors? [7]
- Q.8** (a) Discuss the working principle of clipping circuits. What are the different types of clipping circuits? [7]
 (b) Explain the concepts of static and dynamic resistance in a diode. Derive expressions for both and discuss their significance in diode operation. [7]
- Q.9** Write short notes on *any two* of the following:
 (a) Solar cell
 (b) Basic logic gates, their symbols and truth tables
 (c) Clipping and Clamping circuits
 (d) Transistor biasing techniques